1. Bangladesh Tribute Scene (Using DDA, Bresenham's Mid-Point Line and Circle)

1.1. Screenshot of the output

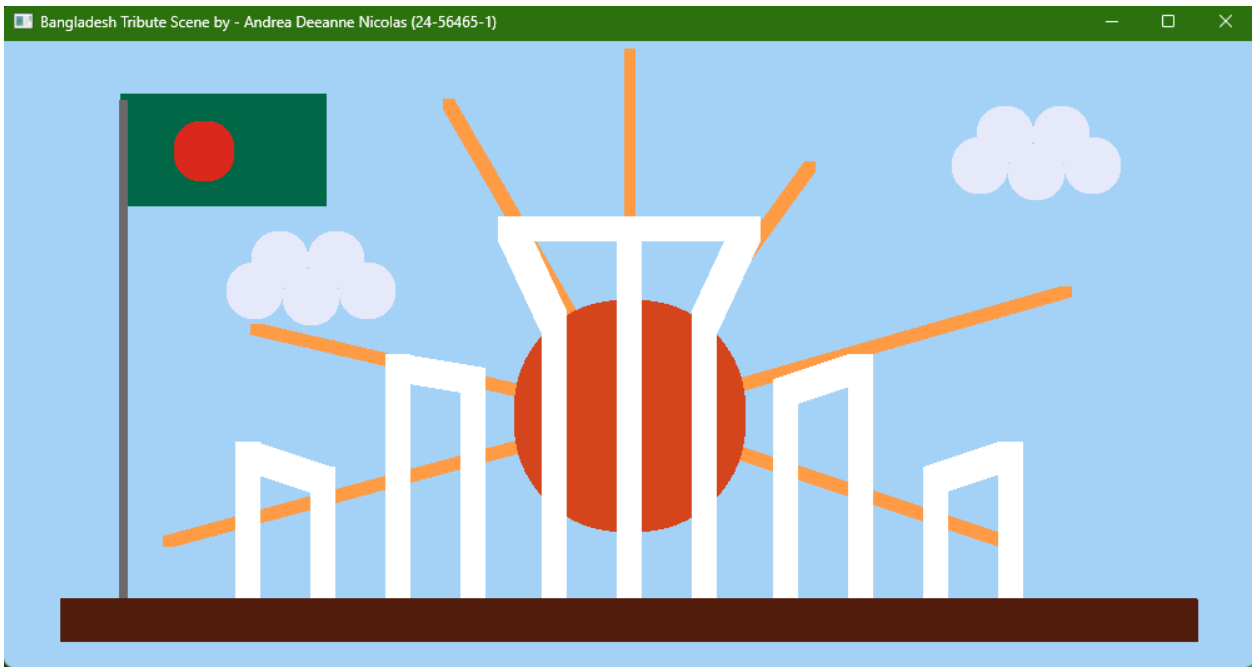

Figure 1: Screenshot of the output of the program.

1.2. Source Code

- GitHub Repository - https://github.com/andrea-nicolas/Bangladesh-Tribute-Scene

```
#include <algorithm>
#include <GL/glut.h>
#include <cmath>
using namespace std;

void DDAline (float x1, float y1, float x2, float y2)
{
    glBegin(GL_POINTS);

    float dx = x2 - x1;
    float dy = y2 - y1;
    float steps;

    if (fabs(dx) > fabs(dy))
    {
        steps = fabs(dx);
    }
    else
    {
```

```
        steps = fabs(dy);
    }

    float delx = dx / steps;
    float dely = dy / steps;

    float x = x1;
    float y = y1;

    for (int i = 1; i <= steps; i++)
    {
        glVertex2f(x,y);
        x += delx;
        y += dely;
    }
    glEnd();
}

void BresenhamLine (float x1, float y1, float x2, float y2)
{
    glBegin(GL_POINTS);

    float dx = x2 - x1;
    float dy = y2 - y1;

    float m = dy / dx;

    float x = x1;
    float y = y1;

    //CASE if not steep
    if (fabs(m) <= 1)
    {
        float d;
        if (m < 0)
        {
            d = 2*fabs(dy) + fabs(dx);
        }
        else
        {
            d = 2*dy - dx;
        }

        for (int i = 0; i <= fabs(dx); i++ )
        {
            glVertex2f(x,y);
            if (d <= 0)
            {
                x++;
                d += 2*fabs(dy);
            }
            else
            {
                if (m < 0) //CASE if line is negative slope
                {
                    y--;
                }
                else //CASE if line is positive slope
                {
                    y++;
                }
                 d += 2*(fabs(dy) - fabs(dx));
                x++;
            }
        }
    }
    //CASE if steep
    else
    {
        float d;
        if (m < 0)
```

```
        {
            d = 2*dx + dy;
        }
        else
        {
            d = 2*dx - dy;
        }

        for (int i = 0; i <= fabs(dy); i++ )
        {
            glVertex2f(x,y);
            if (d <= 0)
            {
                if (m < 0) //CASE if line is negative slope
                {
                    y--;
                }
                else //CASE if line is positive slope
                {
                    y++;
                }
                d += 2*dx;
            }
            else
            {
                if (m < 0) //CASE if line is negative slope
                {
                    y--;
                    d += 2*(dx + dy);
                }
                else //CASE if line is positive slope
                {
                    y++;
                    d += 2*(dx - dy);
                }
                x++;
            }
        }
    }
    glEnd();
}

void BresenhamCircle(int xcenter, int ycenter, int r)
{
    glBegin(GL_POINTS);

    int d = 1 - r;
    int x = 0;
    int y = r;

    while (x <= y)
    {
        glVertex2f(xcenter + x, ycenter + y);
        glVertex2f(xcenter - x, ycenter + y);
        glVertex2f(xcenter + x, ycenter - y);
        glVertex2f(xcenter - x, ycenter - y);

        glVertex2f(xcenter + y, ycenter + x);
        glVertex2f(xcenter - y, ycenter + x);
        glVertex2f(xcenter + y, ycenter - x);
        glVertex2f(xcenter - y, ycenter - x);

        if (d <= 0)
        {
            d += 2*x + 3;
        }
        else
        {
            d += 2*(x - y) + 5;
            y--;
        }
```

```
        x++;
    }

    glEnd();
}

void display()
{
    glClear(GL_COLOR_BUFFER_BIT);

    // --------------------- PART A: Bangladesh Flag ------------------------
    //Rectangle flag frame
    glPointSize(15.0f);
    glColor3f(0.0f, 0.404f, 0.278f);
    BresenhamLine(-400,450,-250,450);
    BresenhamLine(-400,375,-250,375);
    BresenhamLine(-400,375,-400,450);
    BresenhamLine(-250,375,-250,450);
        BresenhamLine(-385,375,-385,450);  //filler
        BresenhamLine(-370,375,-370,450);  //filler
        BresenhamLine(-355,375,-355,450);  //filler
        BresenhamLine(-340,375,-340,450);  //filler
        BresenhamLine(-325,375,-325,450);  //filler
        BresenhamLine(-310,375,-310,450);  //filler
        BresenhamLine(-295,375,-295,450);  //filler
        BresenhamLine(-280,375,-280,450);  //filler
        BresenhamLine(-265,375,-265,450);  //filler


    //Flagpole
    glPointSize(7.0f);
    glColor3f(0.42f, 0.408f, 0.404f);
    DDAline(-405,50,-405,450);

    //Red circle
    glPointSize(8.0f);
    glColor3f(0.855f, 0.161f, 0.11f);
    BresenhamCircle(-340,412,20);
        BresenhamCircle(-340,412,12); //filler
        BresenhamCircle(-340,412,4); //filler

    // --------------------- PART C: Extra Elements ------------------------
    //Sun rays
    glPointSize(9.0f);
    glColor3f(1.0f,0.608f,0.271f);
    DDAline(0,200,0,490);

        //right
        DDAline(0,200,145,400);
        DDAline(0,200,350,300);
        DDAline(0,200,300,100);

        //left
        DDAline(0,200,-145,450);
        DDAline(0,200,-300,270);
        DDAline(0,200,-370,100);

    //Cloud
    glPointSize(5.0f);
    glColor3f(0.898f,0.914f,0.98f);
    BresenhamCircle(-300,300,20);
        BresenhamCircle(-300,300,15); //filler
        BresenhamCircle(-300,300,10); //filler
        BresenhamCircle(-300,300,5); //filler
        BresenhamCircle(-300,300,1); //filler

    BresenhamCircle(-255,295,20);
        BresenhamCircle(-255,295,10); //filler
        BresenhamCircle(-255,295,15); //filler
        BresenhamCircle(-255,295,5); //filler
```

```
    BresenhamCircle(-255,295,1); //filler

BresenhamCircle(-210,300,20);
    BresenhamCircle(-210,300,10); //filler
    BresenhamCircle(-210,300,15); //filler
    BresenhamCircle(-210,300,5); //filler
    BresenhamCircle(-210,300,1); //filler

BresenhamCircle(-280,325,20);
    BresenhamCircle(-280,325,10); //filler
    BresenhamCircle(-280,325,15); //filler
    BresenhamCircle(-280,325,5); //filler
    BresenhamCircle(-280,325,1); //filler

BresenhamCircle(-235,325,20);
    BresenhamCircle(-235,325,10); //filler
    BresenhamCircle(-235,325,15); //filler
    BresenhamCircle(-235,325,5); //filler
    BresenhamCircle(-235,325,1); //filler



BresenhamCircle(280,400,20);
    BresenhamCircle(280,400,10); //filler
    BresenhamCircle(280,400,15); //filler
    BresenhamCircle(280,400,5); //filler
    BresenhamCircle(280,400,1); //filler

BresenhamCircle(325,395,20);
    BresenhamCircle(325,395,10); //filler
    BresenhamCircle(325,395,15); //filler
    BresenhamCircle(325,395,5); //filler
    BresenhamCircle(325,395,1); //filler

BresenhamCircle(370,400,20);
    BresenhamCircle(370,400,10); //filler
    BresenhamCircle(370,400,15); //filler
    BresenhamCircle(370,400,5); //filler
    BresenhamCircle(370,400,1); //filler

BresenhamCircle(300,425,20);
    BresenhamCircle(300,425,10); //filler
    BresenhamCircle(300,425,15); //filler
    BresenhamCircle(300,425,5); //filler
    BresenhamCircle(300,425,1); //filler

BresenhamCircle(345,425,20);
    BresenhamCircle(345,425,10); //filler
    BresenhamCircle(345,425,15); //filler
    BresenhamCircle(345,425,5); //filler
    BresenhamCircle(345,425,1); //filler



// -------------------- PART B: Shaheed Minar ------------------------
//Sun
glPointSize(15.0f);
glColor3f(0.835f,0.271f,0.106f);
BresenhamCircle(0,200,85);
//added extra circles to fill in the sun
    BresenhamCircle(0,200,70); //filler
    BresenhamCircle(0,200,55); //filler
    BresenhamCircle(0,200,40); //filler
    BresenhamCircle(0,200,25); //filler
    BresenhamCircle(0,200,10); //filler

//Third (Centre) Pillar
glPointSize(20.0f);
glColor3f(1.0f, 1.0f, 1.0f);
BresenhamLine(0,62,0,350);
BresenhamLine(60,62,60,275);
```

```
    BresenhamLine(-60,62,-60,275);
    BresenhamLine(-95,350,95,350);
    BresenhamLine(60,275,95,350);
    BresenhamLine(-95,350,-60,275);

    //Second Pillar
    BresenhamLine(-185,62,-185,240);
    BresenhamLine(-125,62,-125,220);
    BresenhamLine(-185,240,-125,230);

    //First Pillar
    BresenhamLine(-245,62,-245,150);
    BresenhamLine(-305,62,-305,170);
    BresenhamLine(-305,170,-245,150);

    //Fourth Pillar
    BresenhamLine(185,62,185,240);
    BresenhamLine(125,62,125,220);
    BresenhamLine(125,220,185,240);

    //Fifth Pillar
    BresenhamLine(245,62,245,150);
    BresenhamLine(305,62,305,170);
    BresenhamLine(245,150,305,170);

    //Base Platform
    glPointSize(10.0f);
    glColor3f(0.322f, 0.11f, 0.051f);
    DDAline(-450,50,450,50);
    DDAline(-450,25,450,25);
    DDAline(-450,25,-450,50);
    DDAline(450,25,450,50);

    glPointSize(25.0f);        //filler
    DDAline(-440,37,440,37);

    glFlush();
}

void init()
{
    glClearColor(0.643f, 0.824f, 0.969f, 0.1f);

    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-500, 500, 0, 500);
}

int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(1000,500);
    glutInitWindowPosition(480,270);
    glutCreateWindow("Bangladesh Tribute Scene by - Andrea Deeanne Nicolas (24-56465-1)");

    init();
    glutDisplayFunc(display);

    glutMainLoop();
    return 0;
}
```

1.3. Details

1.3.1. DDA Algorithm

1.3.1.1. Parts Using DDA Algorithm

a. Flagpole
b. Base Platform (and its filler)
c. Sun rays

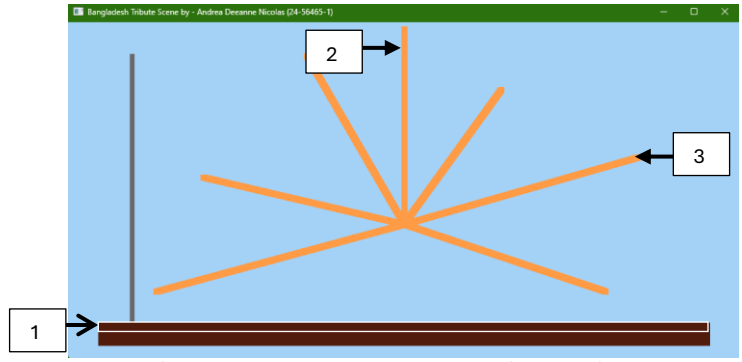

Figure 2: Elements that used DDA Line algorithm.

1.3.1.2. 3 Sample Point Outputs

a. First 5 coordinates generated by the algorithm for Line 1

x	y	round(y)
-449	50	50
-448	50	50
-447	50	50
-446	50	50
-445	50	50

b. First 5 coordinates generated by the algorithm for Line 2

x	y	round(y)
0	201	201
0	202	202
0	203	203
0	204	204
0	204	204

c. First 5 coordinates generated by the algorithm for Line 3

x	y	round(y)
1	200.286	200
2	200.571	201
3	200.857	201
4	201.143	201
5	201.429	201

1.3.2. Bresenham's Mid-Point Line Algorithm

1.3.2.1. Parts Using Bresenham's Mid-Point Line Algorithm

a. Rectangle frame for Bangladesh Flag (and all its fillers)
b. Shaheed Minar pillars

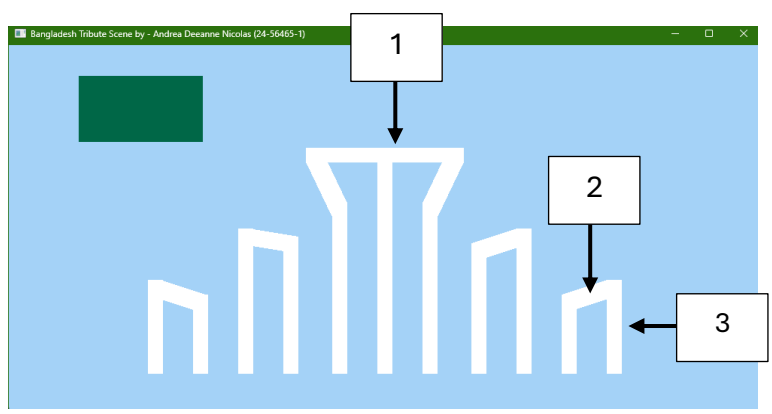

Figure 3: Elements that used Bresenham's Mid-Point Line algorithm.

1.3.2.2. Sample Point Outputs

a. First 5 coordinates generated by the algorithm for Line 1

d	x	y
-190	-94	350
-190	-93	350
-190	-92	350
-190	-91	350
-190	-90	350

b. First 5 coordinates generated by the algorithm for Line 2

d	x	y
-20	246	150
20	247	151
-60	248	151
-20	249	151
20	250	152

c. First 5 coordinates generated by the algorithm for Line 3

d	x	y
-108	305	63
-108	305	64
-108	305	65
-108	305	66
-108	305	67

1.3.3. Parts Using Bresenham's Mid-Point Circle Algorithm

1.3.3.1. Parts Using Bresenham's Mid-Point Circle Algorithm

a. Circle part for Bangladesh Flag (and its fillers)
b. Sun behind Shaheed Minar (and its fillers)
c. Clouds (and its fillers)

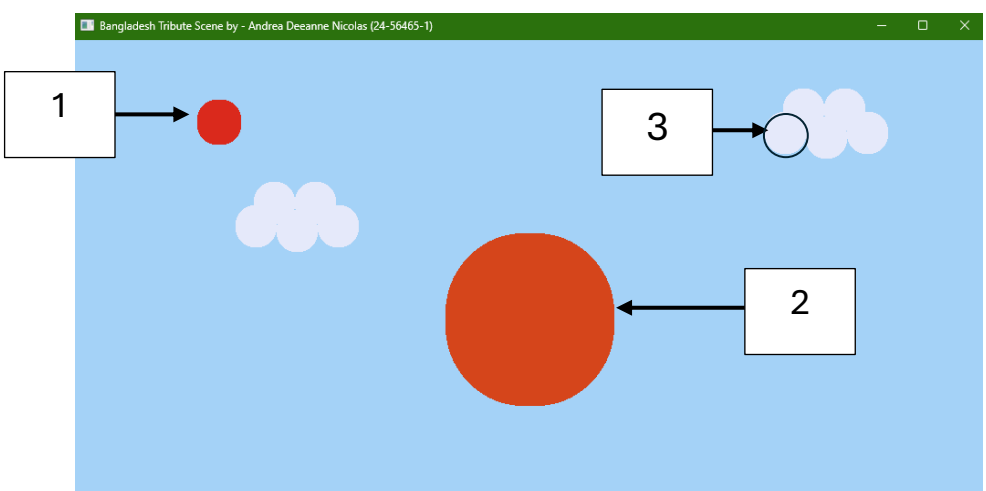

Figure 4: Elements that used Bresenham's Mid-Point Circle algorithm.

1.3.3.2. Sample Point Outputs

a. First 5 coordinates generated by the algorithm for Circle 1

x	y
-340	432
-339	432
-338	432
-337	432
-336	432

b. First 5 coordinates generated by the algorithm for Circle 2

x	y
0	285
1	285
2	285
3	285
4	285

c. First 5 coordinates generated by the algorithm for Circle 3

x	y
280	420
281	420
282	420
283	420
284	420